

Problem Set 1: The Solow–Swan Growth Model

Instructions

Answer all questions. Show all mathematical derivations carefully. Unless otherwise stated, assume

$$0 < \alpha < 1, \quad 0 < s < 1, \quad n \geq 0, \quad g \geq 0, \quad 0 < \delta < 1.$$

Throughout the problem set, define

$$D \equiv (1 + n)(1 + g), \quad b \equiv D - (1 - \delta) = n + g + ng + \delta.$$

The lecture approximation ignores the small second-order term ng , so that

$$b \approx n + g + \delta.$$

Question 1: Equilibrium Dynamics, Steady State, and Balanced Growth [60 marks]

Consider the discrete-time Solow–Swan economy

$$Y_t = F(K_t, A_t L_t) = K_t^\alpha (A_t L_t)^{1-\alpha},$$

where A_t is labour-augmenting technology and L_t is labour. Technology and labour evolve according to

$$A_{t+1} = (1 + g)A_t, \quad L_{t+1} = (1 + n)L_t.$$

Capital accumulation is given by

$$K_{t+1} = (1 - \delta)K_t + I_t.$$

Consumers save a constant fraction s of output:

$$C_t = (1 - s)Y_t, \quad S_t = sY_t,$$

and goods-market clearing implies

$$I_t = S_t.$$

Define capital, output, and consumption per unit of effective labour as

$$k_t \equiv \frac{K_t}{A_t L_t}, \quad y_t \equiv \frac{Y_t}{A_t L_t}, \quad c_t \equiv \frac{C_t}{A_t L_t}.$$

- (a) Derive the intensive-form production function and the law of motion for k_t . Express the law of motion as

$$k_{t+1} = \mathcal{G}(k_t),$$

and derive an expression for $k_{t+1} - k_t$. Explain the economic meaning of the term bk_t . **[15 marks]**

- (b) Solve explicitly for the positive steady state k^* , y^* , and c^* . Then compute the exact and approximate steady-state values when

$$\alpha = \frac{1}{3}, \quad s = 0.24, \quad n = 0.01, \quad g = 0.02, \quad \delta = 0.05.$$

[15 marks]

- (c) Prove that the steady state is unique and globally stable for any $k_0 > 0$. Then derive the local convergence coefficient around k^* :

$$\lambda \equiv \mathcal{G}'(k^*).$$

Compute λ and the half-life of deviations from steady state for the numerical parameters in part (b). **[18 marks]**

- (d) Derive the balanced-growth-path growth rates of K_t , Y_t , C_t , I_t , K_t/L_t , and Y_t/L_t . Then derive the comparative statics of k^* and y^* with respect to s , n , g , and δ . **[12 marks]**

Question 2: Golden Rule, Dynamic Inefficiency, and Saving-Rate Policy [40 marks]

Continue with the Cobb–Douglas Solow economy from Question 1:

$$y = f(k) = k^\alpha.$$

Use the exact break-even investment parameter

$$b = n + g + ng + \delta.$$

Steady-state consumption per unit of effective labour is

$$c^* = (1 - s)f(k^*),$$

where k^* is determined by

$$sf(k^*) = bk^*.$$

- (a) Define the Golden Rule level of capital as the level of k that maximises sustainable steady-state consumption per unit of effective labour:

$$c(k) = f(k) - bk.$$

Derive k_{GR} , y_{GR} , c_{GR} , and the saving rate s_{GR} that decentralises the Golden Rule steady state. [12 marks]

- (b) Using

$$\alpha = \frac{1}{3}, \quad s_0 = 0.24, \quad n = 0.01, \quad g = 0.02, \quad \delta = 0.05,$$

compute k_{GR} , y_{GR} , c_{GR} , and s_{GR} . Compare the initial economy with $s_0 = 0.24$ to the Golden Rule allocation. Is the economy saving too much or too little? [10 marks]

- (c) Suppose the government permanently raises the saving rate from

$$s_0 = 0.24$$

to

$$s_1 = 0.40.$$

Compute the immediate effect on consumption per unit of effective labour, holding k fixed at the old steady state. Then compute the new steady-state values k_1^* , y_1^* , and c_1^* . Does the policy raise long-run consumption relative to the initial steady state? Does it achieve the Golden Rule? [10 marks]

- (d) Derive the local transition dynamics around the new steady state after the saving-rate change. Let

$$\widehat{k}_t \equiv \ln k_t - \ln k_1^*.$$

Show that, locally,

$$\widehat{k}_{t+1} \approx \lambda \widehat{k}_t,$$

where λ is the convergence coefficient. Then derive the approximate growth rate of output per worker during the transition. [**8 marks**]